

# Java Backend Development Learning Syllabus

Prepared for Working Professional

May 22 – August 16, 2025

## Introduction

This syllabus outlines a 12-week learning plan for mastering Java Backend Development with Spring Boot, aligned with the GeeksforGeeks Java Backend Development Live course. It includes a pre-preparation phase (May 22–25, 2025) and daily tasks for Monday–Saturday (10 PM study sessions), with Sundays off. Each day includes topics, tasks, resources, and time estimates.

## Pre-Preparation Phase (May 22–25, 2025)

| Date         | Day      | Topic                                    | Tasks                                                                                                  | Resources                                                                        | Time |
|--------------|----------|------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------|
| May 22, 2025 | Thursday | Java Syntax & Environment Setup          | Install JDK 17, IntelliJ IDEA. Write "Hello World" and programs with variables, loops.                 | <a href="#">Oracle JDK</a> , <a href="#">IntelliJ</a> , <a href="#">GFG Java</a> | 2.5h |
| May 23, 2025 | Friday   | OOP Basics (Classes, Objects)            | Read about classes/objects. Code a Student class with attributes/methods. Practice method overloading. | <a href="#">GFG OOP</a> , <a href="#">Oracle Tutorial</a>                        | 2.5h |
| May 24, 2025 | Saturday | OOP Advanced (Inheritance, Polymorphism) | Implement inheritance (Person → Student). Practice polymorphism with method overriding.                | <a href="#">GFG Inheritance</a> , <a href="#">FreeCodeCamp Java (YouTube)</a>    | 2.5h |
| May 25, 2025 | Sunday   | Exception Handling & Review              | Read about try-catch, finally. Code a program handling ArithmeticException. Review prep concepts.      | <a href="#">GFG Exceptions</a> , <a href="#">HackerRank Java</a>                 | 2h   |

## Main Course Phase (May 26–August 16, 2025)

| Date         | Day       | Week/Session | Topic                           | Tasks                                                                            | Time                           |
|--------------|-----------|--------------|---------------------------------|----------------------------------------------------------------------------------|--------------------------------|
| May 26, 2025 | Monday    | W1/S1        | Java Architecture & OOP Basics  | Read JDK, JRE, JVM. Code a class with encapsulation.                             | GFG Java Architecture          |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| May 27, 2025 | Tuesday   | W1/S1        | OOP (Inheritance, Polymorphism) | Implement inheritance and polymorphism (Animal → Dog). Solve 2 GFG problems.     | GFG Inheritance                |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| May 28, 2025 | Wednesday | W1/S1        | Abstraction & Encapsulation     | Code an abstract class and interface. Practice encapsulation with BankAccount.   | GFG Abstraction                |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| May 29, 2025 | Thursday  | W1/S2        | Abstract Classes vs. Interfaces | Compare abstract classes vs. interfaces with code examples. Solve 1 GFG problem. | GFG Abstract vs Interface      |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| May 30, 2025 | Friday    | W1/S2        | Exception Handling              | Write a program with try-catch, throw, finally. Handle NullPointerException.     | GFG Exception                  |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| May 31, 2025 | Saturday  | W1/S2        | Exception Propagation           | Code a program to demonstrate exception propagation. Review Week 1.              | GFG Exception Propagation      |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| Jun 2, 2025  | Monday    | W2/S3        | Lambda Expressions              | Read about lambdas. Write a lambda for sorting a list. Use Predicate.            | GFG Lambda, Baeldung Lambda    |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| Jun 3, 2025  | Tuesday   | W2/S3        | Functional Interfaces           | Use Consumer, Function interfaces. Solve 1 GFG problem.                          | Baeldung Functional Interfaces |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| Jun 4, 2025  | Wednesday | W2/S3        | Java Collections                | Practice ArrayList, HashSet. Code a program to store student data.               | GFG Collections                |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| Jun 5, 2025  | Thursday  | W2/S4        | Generics                        | Read about generics. Code a generic class/method.                                | GFG Generics                   |
| 1.5h         |           |              |                                 |                                                                                  |                                |
| Jun 6, 2025  | Friday    | W2/S4        | Streams API                     | Practice streams (filter, map). Solve 1 GFG problem.                             | Baeldung Streams               |

| Date             | Day           | Week/Session | Topic                   | Tasks                                                                         | Time                                                    |
|------------------|---------------|--------------|-------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------|
| 1.5h<br>Jun 2025 | 7, Saturday   | W2/S4        | HashMap                 | Code a program using HashMap. Review Week 2.                                  | <a href="#">GFG HashMap</a>                             |
| 1.5h<br>Jun 2025 | 9, Monday     | W3/S5        | Multithreading Basics   | Read about threads. Code a simple thread using Runnable.                      | <a href="#">GFG Multithreading</a>                      |
| 1.5h<br>Jun 2025 | 10, Tuesday   | W3/S5        | Thread Synchronization  | Implement synchronized methods. Solve 1 threading problem.                    | <a href="#">Baeldung Thread Safety</a>                  |
| 1.5h<br>Jun 2025 | 11, Wednesday | W3/S5        | Thread Pools            | Use ExecutorService for thread management.                                    | <a href="#">Baeldung ExecutorService</a>                |
| 1.5h<br>Jun 2025 | 12, Thursday  | W3/S6        | Maven Basics            | Set up a Maven project. Explore POM.xml.                                      | <a href="#">GFG Maven</a>                               |
| 1.5h<br>Jun 2025 | 13, Friday    | W3/S6        | Maven Plugins           | Add a plugin to POM.xml. Build a project.                                     | <a href="#">Maven Docs</a>                              |
| 1.5h<br>Jun 2025 | 14, Saturday  | W3/S6        | Maven Review            | Create a new Maven project with archetype. Review Week 3.                     | Maven Docs                                              |
| 1.5h<br>Jun 2025 | 16, Monday    | W4/S7        | Spring Framework Basics | Read about DI, IoC. Code a simple Spring app with XML config.                 | <a href="#">Baeldung Spring</a>                         |
| 1.5h<br>Jun 2025 | 17, Tuesday   | W4/S7        | Spring Beans            | Configure beans using annotations. Explore Bean scopes.                       | <a href="#">GFG Spring Beans</a>                        |
| 1.5h<br>Jun 2025 | 18, Wednesday | W4/S8        | Spring Boot Setup       | Set up a Spring Boot project with Spring Initializr. Run a "Hello World" API. | <a href="#">Spring Initializr, Baeldung Spring Boot</a> |
| 1.5h<br>Jun 2025 | 19, Thursday  | W4/S8        | Spring Boot Annotations | Use @RestController, @Autowired. Build a simple API.                          | Baeldung Spring Boot                                    |
| 1.5h<br>Jun 2025 | 20, Friday    | W4/S8        | Spring Boot Deployment  | Package and run a Spring Boot app locally. Review Week 4.                     | Baeldung Deployment                                     |

| Date                 | Day       | Week/Session | Topic                   | Tasks                                                                | Time                                   |
|----------------------|-----------|--------------|-------------------------|----------------------------------------------------------------------|----------------------------------------|
| 1.5h<br>Jun 23, 2025 | Monday    | W5/S9        | Spring MVC Basics       | Read about MVC. Set up a Spring MVC project.                         | <a href="#">GFG Spring MVC</a>         |
| 1.5h<br>Jun 24, 2025 | Tuesday   | W5/S9        | REST API Basics         | Code a GET/POST API for a Book entity.                               | <a href="#">Baeldung REST</a>          |
| 1.5h<br>Jun 25, 2025 | Wednesday | W5/S9        | HTTP Methods            | Implement DELETE/PUT APIs. Solve 1 REST problem.                     | <a href="#">Baeldung REST</a>          |
| 1.5h<br>Jun 26, 2025 | Thursday  | W5/S10       | Spring AOP              | Read about AOP. Implement a logging aspect.                          | <a href="#">Baeldung AOP</a>           |
| 1.5h<br>Jun 27, 2025 | Friday    | W5/S10       | Lombok                  | Use Lombok annotations (@Getter, @Setter). Integrate in Spring Boot. | <a href="#">Baeldung Lombok</a>        |
| 1.5h<br>Jun 28, 2025 | Saturday  | W5/S10       | Spring Logging          | Set up logging in Spring Boot. Review Week 5.                        | <a href="#">Baeldung Logging</a>       |
| 1.5h<br>Jun 30, 2025 | Monday    | W6/S11       | JDBC Basics             | Read about JDBC. Write a program to connect to MySQL.                | <a href="#">GFG JDBC</a>               |
| 1.5h<br>Jul 1, 2025  | Tuesday   | W6/S11       | Spring JDBC Template    | Use JDBC Template for CRUD operations.                               | <a href="#">Baeldung JDBC Template</a> |
| 1.5h<br>Jul 2, 2025  | Wednesday | W6/S11       | Connection Pooling      | Configure HikariCP in Spring Boot.                                   | <a href="#">Baeldung JDBC Template</a> |
| 1.5h<br>Jul 3, 2025  | Thursday  | W6/S12       | Hibernate Basics        | Read about JPA/Hibernate. Create an Entity class.                    | <a href="#">Baeldung JPA</a>           |
| 1.5h<br>Jul 4, 2025  | Friday    | W6/S12       | JPA CRUD                | Implement CRUD with JPA Repository.                                  | <a href="#">Baeldung JPA</a>           |
| 1.5h<br>Jul 5, 2025  | Saturday  | W6/S12       | JPA Relationships       | Code One-to-Many relationship. Review Week 6.                        | <a href="#">Baeldung JPA</a>           |
| 1.5h<br>Jul 7, 2025  | Monday    | W7/S13       | Digital Library Project | Set up project. Design database schema (Book, User).                 | <a href="#">Baeldung CRUD</a>          |

| Date                 | Day       | Week/Session | Topic                   | Tasks                                                    | Time                                     |
|----------------------|-----------|--------------|-------------------------|----------------------------------------------------------|------------------------------------------|
| 2h<br>Jul 8, 2025    | Tuesday   | W7/S13       | Project APIs            | Implement Book CRUD APIs.                                | Baeldung<br>CRUD                         |
| 2h<br>Jul 9, 2025    | Wednesday | W7/S13       | Project Flowchart       | Create project flowchart.<br>Test APIs with Postman.     | Postman:<br><a href="#">Postman</a>      |
| 2h<br>Jul 10, 2025   | Thursday  | W7/S14       | Project Features        | Add user management APIs.                                | Baeldung<br>CRUD                         |
| 2h<br>Jul 11, 2025   | Friday    | W7/S14       | Project Data Modeling   | Implement data modeling (relationships).                 | Baeldung<br>JPA                          |
| 2h<br>Jul 12, 2025   | Saturday  | W7/S14       | Project Review          | Test and refine project. Review Week 7.                  | Postman                                  |
| 2h<br>Jul 14, 2025   | Monday    | W8/S15       | JUnit Testing           | Write JUnit tests for repository layer.                  | <a href="#">Baeldung JUnit</a>           |
| 1.5h<br>Jul 15, 2025 | Tuesday   | W8/S15       | Mockito Testing         | Use Mockito for service layer testing.                   | <a href="#">Baeldung Mockito</a>         |
| 1.5h<br>Jul 16, 2025 | Wednesday | W8/S15       | MockMvc Testing         | Test controllers with MockMvc.                           | Baeldung<br>Mockito                      |
| 1.5h<br>Jul 17, 2025 | Thursday  | W8/S16       | Redis Basics            | Install Redis locally. Cache a simple key-value pair.    | <a href="#">Redis Do</a>                 |
| 1.5h<br>Jul 18, 2025 | Friday    | W8/S16       | Redis in Spring Boot    | Integrate Redis with Spring Boot.                        | <a href="#">Baeldung Redis</a>           |
| 1.5h<br>Jul 19, 2025 | Saturday  | W8/S16       | Caching Review          | Test caching performance. Review Week 8.                 | Baeldung<br>Redis                        |
| 1.5h<br>Jul 21, 2025 | Monday    | W9/S17       | Spring Security Basics  | Read about Spring Security. Set up basic authentication. | <a href="#">Baeldung Spring Security</a> |
| 1.5h<br>Jul 22, 2025 | Tuesday   | W9/S17       | Authorization           | Implement role-based authorization.                      | Baeldung<br>Spring Security              |
| 1.5h<br>Jul 23, 2025 | Wednesday | W9/S17       | Database Authentication | Configure database user authentication.                  | Baeldung<br>Spring Security              |
| 1.5h                 |           |              |                         |                                                          |                                          |

| Date                                     | Day       | Week/Session | Topic                 | Tasks                                                       | Time                                                  |
|------------------------------------------|-----------|--------------|-----------------------|-------------------------------------------------------------|-------------------------------------------------------|
| Jul 24, 2025                             | Thursday  | W9/S18       | Project: Security     | Testing                                                     | Integrate Spring Security and JWT in Digital Library. |
| Baeldung Spring Security, Baeldung JUnit | 2h        |              |                       |                                                             |                                                       |
| Jul 25, 2025                             | Friday    | W9/S18       | Project: Caching      | Add Redis caching to Digital Library.                       | Baeldung Redis                                        |
| 2h                                       |           |              |                       |                                                             |                                                       |
| Jul 26, 2025                             | Saturday  | W9/S18       | Project Review        | Add features (e.g., search). Review Week 9.                 | Postman                                               |
| 2h                                       |           |              |                       |                                                             |                                                       |
| Jul 28, 2025                             | Monday    | W10/S19      | OAuth2 Basics         | Read about OAuth2. Explore GitHub OAuth flow.               | <a href="#">Baeldung OAuth2</a>                       |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Jul 29, 2025                             | Tuesday   | W10/S19      | OAuth2 Implementation | Implement GitHub OAuth2 in Spring Boot.                     | Baeldung OAuth2                                       |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Jul 30, 2025                             | Wednesday | W10/S20      | Microservices Basics  | Read about microservices vs. monolithic. Set up 2 services. | <a href="#">Baeldung Microservices</a>                |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Jul 31, 2025                             | Thursday  | W10/S20      | Feign Client          | Use Feign for inter-service communication.                  | <a href="#">Baeldung Feign</a>                        |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Aug 1, 2025                              | Friday    | W10/S20      | Eureka Server         | Set up Eureka for service discovery.                        | <a href="#">Baeldung Eureka</a>                       |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Aug 2, 2025                              | Saturday  | W10/S20      | Microservices Review  | Test microservices. Review Week 10.                         | Baeldung Microservices                                |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Aug 4, 2025                              | Monday    | W11/S21      | Kafka Basics          | Install Kafka locally. Send/receive messages.               | <a href="#">Kafka Quickstart</a>                      |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Aug 5, 2025                              | Tuesday   | W11/S21      | Kafka in Spring Boot  | Integrate Kafka with Spring Boot.                           | <a href="#">Baeldung Kafka</a>                        |
| 1.5h                                     |           |              |                       |                                                             |                                                       |
| Aug 6, 2025                              | Wednesday | W11/S21      | Spring Cloud          | Set up Spring Cloud and Hystrix.                            | <a href="#">Baeldung Spring Cloud</a>                 |

| Date             | Day           | Week/Session | Topic                | Tasks                                                                | Time                                  |
|------------------|---------------|--------------|----------------------|----------------------------------------------------------------------|---------------------------------------|
| 1.5h<br>Aug 2025 | 7, Thursday   | W11/S22      | E-Wallet Project     | Set up E-Wallet project. Design database schema.                     | Baeldung<br>Microser-<br>vices        |
| 2h<br>Aug 2025   | 8, Friday     | W11/S22      | Project APIs         | Implement payment APIs.                                              | Baeldung<br>CRUD                      |
| 2h<br>Aug 2025   | 9, Saturday   | W11/S22      | Project Architecture | Create microservices archi-<br>tecture. Review Week 11.              | Baeldung<br>Microser-<br>vices        |
| 2h<br>Aug 2025   | 11, Monday    | W12/S23      | E-Wallet Continued   | Add Spring Cloud Gateway<br>for routing.                             | <a href="#">Baeldung<br/>Gateway</a>  |
| 2h<br>Aug 2025   | 12, Tuesday   | W12/S23      | Project Features     | Implement additional fea-<br>tures (e.g., transaction his-<br>tory). | Baeldung<br>CRUD                      |
| 2h<br>Aug 2025   | 13, Wednesday | W12/S23      | Project Testing      | Test APIs with Postman.                                              | Postman                               |
| 2h<br>Aug 2025   | 14, Thursday  | W12/S24      | CI/CD Setup          | Set up GitHub Actions for<br>CI/CD.                                  | <a href="#">GitHub<br/>Actions</a>    |
| 2h<br>Aug 2025   | 15, Friday    | W12/S24      | Cloud Deployment     | Deploy E-Wallet to Heroku.                                           | <a href="#">Baeldung<br/>Heroku</a>   |
| 2h<br>Aug 2025   | 16, Saturday  | W12/S24      | Monitoring           | Set up basic monitoring<br>with Prometheus. Review<br>course.        | <a href="#">Baeldung<br/>Prometho</a> |
| 2h               |               |              |                      |                                                                      |                                       |

## Tracking Progress

- Use a Google Sheet to log daily task completion, notes, and challenges.
- Create a GitHub repository for code (practice, projects).
- Solve 1–2 problems daily on [GeeksforGeeks Practice](#) or HackerRank.
- Review weekly with Grok: ask for explanations, debugging, or quizzes.